Assignment No. 03 — Section 4: Results Tables

Team Members: Rizwana Mubarak (23F-6055) & Abdul Rehman (23F-6075)

4.1 Table 1 – Data

Metric	Value
Train / validation / test rows	78,799 / 8,755 / 21,892
Minimum / maximum signal value	0.0000 / 1.0000
Rows per class, train file (N, S, V, F, Q)	72,471, 2,223, 5,788, 641, 6,431 (Total: 87,554)
Rows per class, test file (N, S, V, F, Q)	18,118, 556, 1,448, 162, 1,608 (Total: 21,892)

4.2 Table 2 – Network and training

Metric	Value
Trainable parameters	32,645 (Layer 1: 24,064 | Layer 2: 8,256 | Layer 3: 325)
Initial loss (before training)	1.7202 (Theoretical log 5 = 1.6094)
Gradient check: max difference for W[3] / b[3]	2.98e-08 / 5.96e-08
Chosen alpha	0.5
Best epoch (lowest validation loss)	25
Training / validation loss at best epoch	0.0535 / 0.0716
Training / validation accuracy at best epoch	98.33% / 97.84%
Training time and hardware	36.2s (Colab CPU)

4.3 Table 3 – Learning-rate choice

alpha	Lowest validation loss	Validation accuracy at that epoch	Best epoch
0.01	0.1618	95.79%	30
0.1	0.0867	97.59%	28
0.5	0.0716	97.84%	25

4.4 Table 4 – Test results

Per-class accuracy	N: 99.56%	S: 69.24%	V: 91.78%	F: 54.32%	Q: 97.76%
Overall test metrics	Test accuracy: 97.81%		Balanced accuracy: 82.53%		Always-"N": 82.76%

Full 5 x 5 Test Confusion Matrix (Rows = True Class, Columns = Predicted Class)

True \ Predicted	N (0)	S (1)	V (2)	F (3)	Q (4)	Row Total
N – normal beat (0)	18,039	37	28	1	13	18,118
S – supraventricular (1)	163	385	6	0	2	556
V – ventricular (2)	94	8	1,329	14	3	1,448
F – fusion beat (3)	54	1	19	88	0	162
Q – unknown / paced (4)	25	0	11	0	1,572	1,608